

Ultra-efficient frequency conversion in quasi-phase-matched lithium niobate microrings

JIA-YANG CHEN,^{1,2}  ZHAO-HUI MA,^{1,2} YONG MENG SUA,^{1,2}  ZHAN LI,^{1,2}
CHAO TANG,^{1,2} AND YU-PING HUANG^{1,2,*}

¹Department of Physics, Stevens Institute of Technology, 1 Castle Point Terrace, Hoboken, New Jersey 07030, USA

²Center for Quantum Science and Engineering, Stevens Institute of Technology, 1 Castle Point Terrace, Hoboken, New Jersey 07030, USA

*Corresponding author: yuping.huang@stevens.edu

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We demonstrate quasi-phase-matched frequency conversion in a chip-integrated lithium niobate microring resonator, whose normalized efficiency reaches 230,000%/W or 10⁻⁶ per single photon. © 2019 Optical Society of America under the terms of the [OSA Open Access Publishing Agreement](#)

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Efficient nonlinear optical functions on chip will escalate the utilities and performance of photonic integrated circuits for a breadth of classical and quantum applications. Combining exceptional linear and electrical optical properties with strong second-order nonlinearities, thin-film lithium niobate on insulator (LNOI) has emerged as a powerful nanophotonic platform that integrates a series of linear and nonlinear functions. Thus far, remarkable progress has been made to realize a variety of linear and nonlinear LNOI circuits with outstanding performance [1–3]. To fully capitalize on the LNOI's potential for nonlinear optics, however, it requires using its largest nonlinear tensor elements, with the simultaneous realizations of phase matching, long interaction length, tight mode confinement, and good mode overlapping for all light waves. Previous demonstrations [2–7] fall short on one or more such requirements, with the highest frequency conversion efficiency capped on 10,000%/W [3].

Here, we demonstrate a doubly resonant, periodic-poled LNOI microcavity that meets all the stringent requirements to realize ultra-efficient nonlinearities. The absolute efficiency reaches 1.3 % with only 5.6 μ W pump power, which amounts to a normalized conversion efficiency of 230,000%/W. This result represents a 23 times increase from the state of the art [3], and by far the highest among all integrated photonic platforms [2–9]. This high efficiency implies a single-photon nonlinearity of 10⁻⁶, which marks a 20 times improvement over the highest reported value [10]. A comparison of the present work with the highest optical nonlinearities demonstrated on various platforms is summarized in Table 1. The current high efficiency is achieved despite under optimized parameters and a moderate loaded cavity Q_L of 3.7×10^5 . Upon optimization and by further increasing the Q_L , such as demonstrated in [1], the nonlinearity will reach

single-photon level, which will allow deterministic entanglement generation [12], and control-NOT gate for single photons [13], to bring scalable, nonlinear-optical quantum computing into a reality.

The microcavity is shown in Fig. 1(a), which is in a racetrack shape with one arm of a 300 μ m straight waveguide periodically poled to achieve quasi-phase-matching between fundamental cavity modes. All modes are quasi-transverse-electric (TE₀₀) so as to utilize the largest nonlinear coefficient d_{33} offered by the present X-cut LNOI. Enabled by precise and uniform periodical poling [6], the waveguide cross sections are carefully fabricated to tightly confine and optimally overlap all cavity modes. The simulated quasi-TE mode profiles are shown in the insets of Figs. 1(b) and 1(d), whose overlap exceeds 90%. To achieve the double resonance, the chip is placed on a temperature-varying holder so that resonance mismatch between the interaction modes is compensable by using their different thermal-optic responses [7].

The entire device is fabricated on a X-cut magnesium-doped LNOI wafer (NANOLN Inc.), with a 500 nm thick LN thin film bonded on a 2 μ m silicon dioxide layer above a silicon substrate. First, a periodically poled region is created with a period of 4 μ m using a similar process in [6]; see Fig. 1(a). Then, standard electron beam lithography is used to define the racetrack microcavity structure with a top width of 1.8 μ m. Finally, ion-milling [6] is applied to cut the structure with loaded $Q_L \approx 3.7 \times 10^5$

Table 1. Comparison of Optical Nonlinearities Demonstrated in Nanophotonic Resonators^a

Material	Q_L	$\eta\%$ at $P = 35 \mu\text{W}$	$\eta_s(\times 10^{-7})$
Present	3.7×10^5	5.9	9.7
LN [3]	9.3×10^6	0.35	0.017
LN [4]	6.5×10^5	3×10^{-3}	2.2×10^{-3}
LN [5]	1.4×10^5	0.05	0.17
AlN [8]	5.9×10^5	0.59	0.45
GaAs [9]	2.4×10^5	3.5×10^{-3}	3.9×10^{-3}
Si ₃ N ₄ [11]	3.0×10^5	0.34	7.4×10^{-4}

^a η is the conversion efficiency scaled to 35 μ W pump power. η_s is the single photon nonlinearity defined as $\eta E_{\text{ph}}/\tau$ [10], with E_{ph} being the pump's single-photon energy, and τ the cavity lifetime. For Si₃N₄, $\eta_s = \eta(E_{\text{ph}}/\tau)^2$, as it scales quadratically in power by virtue of $\chi^{(3)}$.

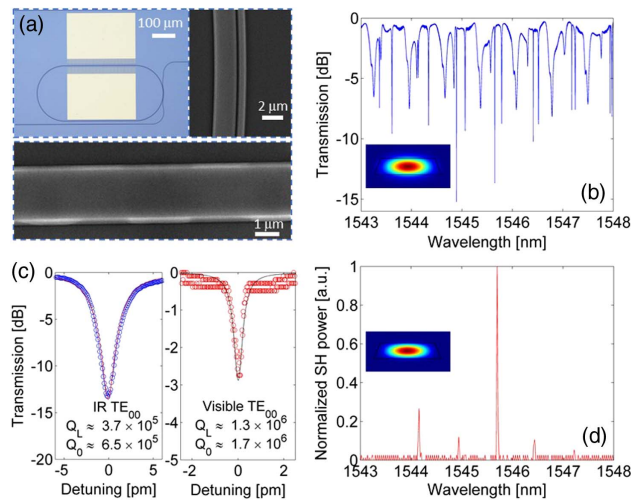


Fig. 1. Device design and characterization. (a) Microscope image (top left) of a patterned device. The scanning electron microscopy images of the coupling region (top right) and poling region (bottom). (b) The measured infrared spectrum for the racetrack microcavity. (c) Zoom-in spectra of infrared and visible TE₀₀ modes. (d) SHG spectrum by sweeping the pump laser from 1543 to 1548 nm. The insets in (b) and (d) are simulated mode profiles for 1545.6 nm and 772.8 nm, respectively.

(IR mode), as shown in Fig. 1(c). The key is to ensure uniform poling and minimally affected sidewall roughness, which is shown in Fig. 1(a). Also shown in the same figure is a pulley waveguide (top width 400 nm) as the ring-bus waveguide coupler to attain good coupling for both IR (95%) and visible light (50%).

To measure the nonlinear efficiency, a tunable laser (Santec 550) is applied through a polarization controller to pump second-harmonic generation (SHG). Two tapered fibers (OZ optics) are used for the input and output couplings with measured losses of 5.5 dB per facet at 1545.6 nm and 8.5 dB per facet for 772.8 nm, respectively. By sweeping the infrared laser and fine tuning the device's temperature, a quasi-phase-matched SHG is achieved for optimal resonance modes at 1545.560 nm and its second-harmonic (SH) at 772.780 nm at 34.5°C. With 20.9 μW pump power in the input fiber, 10.2 nW SH power is collected in the output fiber. Accounting for the coupling loss, the input pump and generated SH powers in the bus waveguide are estimated to be 5.6 μW and 72.2 nW, respectively. This corresponds to a normalized SHG efficiency of 230,000%/W. The measured SH power as a function of the pump power is plotted in Fig. 2(a), which exhibits a quadratic response in the low-power regime. For pump power above 7 μW, thermal effects start to spoil the double-resonance condition and reduce the conversion efficiency. Here, to recover the double resonance in the high-power regime, the temperature sweeping is applied to maximize the SH power. For 35.3 μW pump power, 2.1 μW SH power is created, which amounts to a 6% absolute conversion efficiency. It could be further improved by overly coupling the SH mode while maintaining critical coupling for the pump as its power increases [14]. Figure 2(b) plots the temperature dependency of the SHG

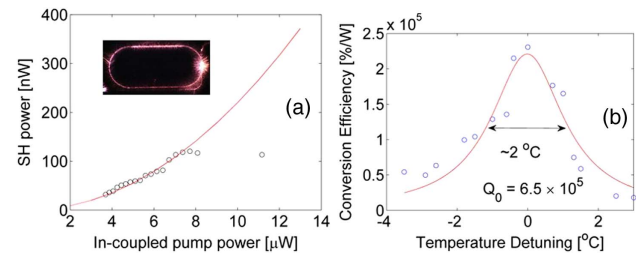


Fig. 2. (a),(b) Pump power and temperature dependency of the generated SH power, respectively, with quadratic and Lorentzian fitting. The inset shows a microscope image of the scattered SH light.

efficiency for the optimized cavity mode (around 1545.6 nm), showing a tolerance of 2°C.

In conclusion, we have demonstrated SHG with an ultra-high efficiency of 230,000%/W or 10^{-6} per photon. This result marks a new height of optical nonlinearities realized in on-chip devices, and may accelerate the development of many scalable quantum and classical applications, particularly those based on LNOI. Several immediate applications include quantum light generation, wavelength conversion, and parametric amplification.

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